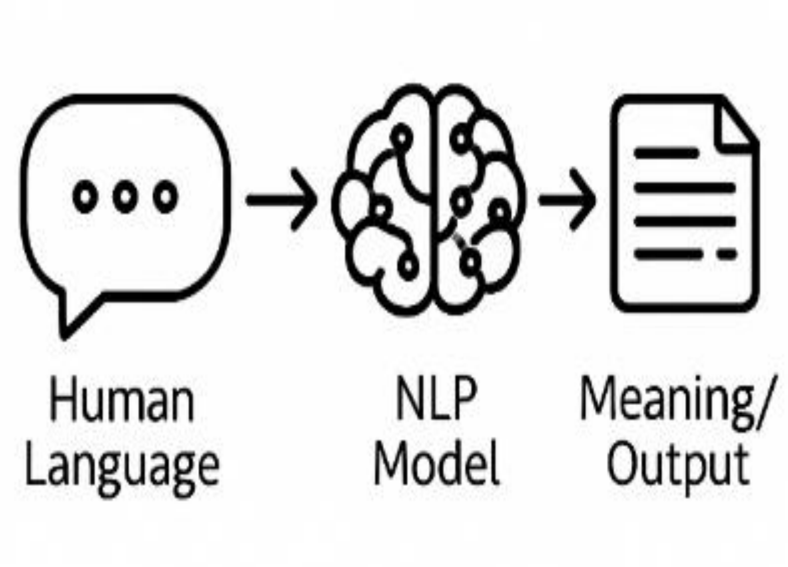


NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

MODULE 5

Syntactic analysis

Dr. Reya Sharma

Assistant Professor

Dept. of Analytics, SCOPE, VIT

Ambiguities in Syntactic Parsing

- Parsing = recognizing an input string + assigning a structure to it.
- Syntactic parsing = recognizing a sentence + assigning a syntactic structure.

Ambiguity

- Ambiguity is perhaps the most serious problem faced by syntactic parsers.
- Structural ambiguity, which arises from many commonly used rules in phrase-structure grammars.

Ambiguities in Syntactic Parsing

- Structural ambiguity occurs when the grammar can assign more than one parse to a sentence.

"One morning I shot an elephant in my pajamas."

- Groucho Marx's well-known line as Captain Spaulding in Animal Crackers is ambiguous because the phrase in my pajamas can be part of the NP headed by elephant or a part of the verb phrase headed by shot."

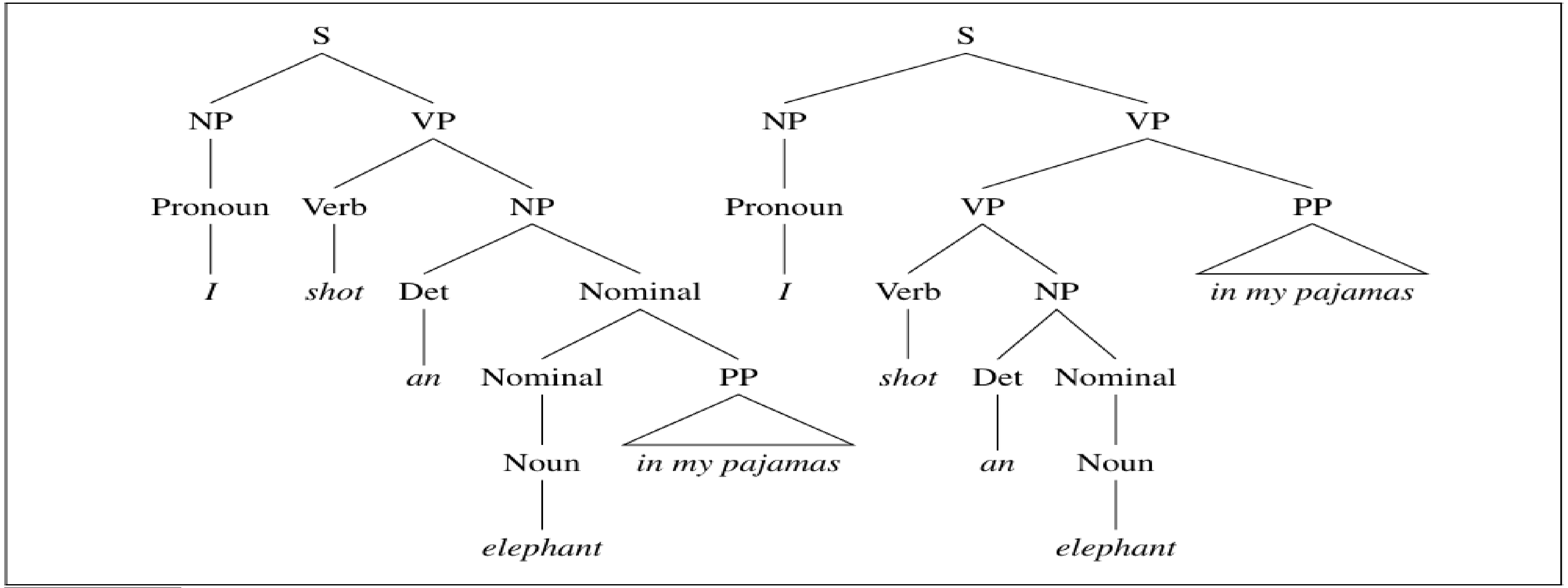
Ambiguities in Syntactic Parsing

Grammar	Lexicon
$S \rightarrow NP\ VP$	$Det \rightarrow that \mid this \mid the \mid a$
$S \rightarrow Aux\ NP\ VP$	$Noun \rightarrow book \mid flight \mid meal \mid money$
$S \rightarrow VP$	$Verb \rightarrow book \mid include \mid prefer$
$NP \rightarrow Pronoun$	$Pronoun \rightarrow I \mid she \mid me$
$NP \rightarrow Proper\text{-}Noun$	$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$NP \rightarrow Det\ Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	$Preposition \rightarrow from \mid to \mid on \mid near \mid through$
$Nominal \rightarrow Nominal\ Noun$	
$Nominal \rightarrow Nominal\ PP$	
$VP \rightarrow Verb$	
$VP \rightarrow Verb\ NP$	
$VP \rightarrow Verb\ NP\ PP$	
$VP \rightarrow Verb\ PP$	
$VP \rightarrow VP\ PP$	
$PP \rightarrow Preposition\ NP$	

The $\mathscr{L}_1$ miniature English grammar and lexicon.

Ambiguities in Syntactic Parsing

Two parse trees for an ambiguous sentence.



The parse on the left corresponds to the humorous reading in which the elephant is in the pajamas, the parse on the right corresponds to the reading in which Captain Spaulding did the shooting in his pajamas.

Ambiguities in Syntactic Parsing

- Structural ambiguity, appropriately enough, comes in many forms.
- Two common kinds of ambiguity are ***attachment ambiguity*** and ***coordination ambiguity***.

i. Attachment Ambiguity

- A sentence has an attachment ambiguity if a particular constituent can be attached to the parse tree at more than one place.
- Example: The Groucho Marx sentence is an example of PP-attachment ambiguity.

Ambiguities in Syntactic Parsing

ii. Coordination Ambiguity

- In coordination ambiguity different sets of phrases can be conjoined by a conjunction like *and*.

  For example:

- The phrase **old men and women** can be bracketed as:
- [old [men and women]], referring to old men and old women,
- or as [old men] and [women], in which case it is only the men who are old.

Ambiguities in Syntactic Parsing

- The fact that there are *many grammatically correct but semantically unreasonable parses* for *naturally occurring sentences* is an irksome *problem that affects all parsers*.
- Ultimately, most *natural language processing systems* need to be able to *choose a* ***single correct parse*** from the *multitude of possible parses* through a process of **syntactic disambiguation**.
- Effective disambiguation algorithms require statistical, semantic, and contextual knowledge sources that vary in how well they can be integrated into parsing algorithms.

Context Free Grammar

- The most widely used formal system for modeling constituent structure in English and other natural languages is the **Context-Free Grammar**, or **CFG.**
- A context-free grammar consists of a **set of rules or productions**, each of which expresses the ways that symbols of the language can be grouped and ordered together, and a lexicon of words and symbols.

Context Free Grammar

- For example, the following productions express that an NP (or noun phrase) can be composed of either a Proper Noun or a determiner (Det) followed by a Nominal; a Nominal in turn can consist of one or more Nouns.

$$\begin{aligned} NP &\rightarrow Det\ Nominal \\ NP &\rightarrow ProperNoun \\ Nominal &\rightarrow Noun \mid Nominal\ Noun \end{aligned}$$

- Context-free rules can be hierarchically embedded, so we can combine the previous rules with others, like the following, that *express facts about the lexicon*:

$$\begin{aligned} Det &\rightarrow a \\ Det &\rightarrow the \\ Noun &\rightarrow flight \end{aligned}$$

Context Free Grammar

- Two classes of symbols in a CFG:

i. **Terminal symbols**
   - Represent actual words in the language.
   - Example: “the”, “nightclub”.
   - The lexicon is the set of rules that introduce these terminal symbols.

ii. **Non-terminal symbols**
   - Express abstractions or generalizations over terminals.
   - Represent linguistic categories (e.g., *Noun Phrase (NP)*, *Verb Phrase (VP)*).

Context Free Grammar

- Structure of a CFG rule:
  - Left side of the rule (LHS) → a single non-terminal.
  - Right side of the rule (RHS) → an ordered list of one or more terminals and/or non-terminals.
- Lexicon's role:
  - Maps words **(terminals)** to their lexical categories / parts of speech **(non-terminals)**.
  - Example:

    Det → the

    N → nightclub

Context Free Grammar

- Viewing a CFG as a generator, we can read **the → arrow** as "*rewrite the symbol on the left* with the *string of symbols on the right*".

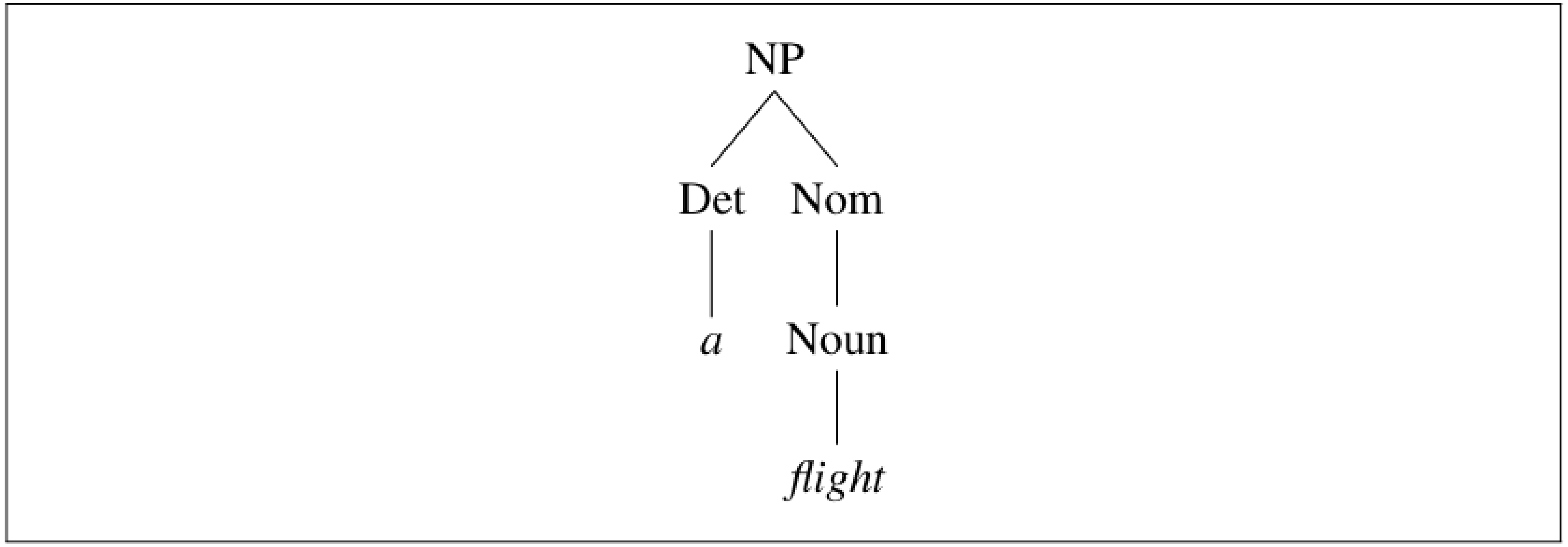
$$
\begin{aligned}
NP &\rightarrow Det\ Nominal \\
NP &\rightarrow ProperNoun \\
Nominal &\rightarrow Noun \mid Nominal\ Noun
\end{aligned}
$$

So starting from the symbol:	*NP*
we can use our first rule to rewrite *NP* as:	*Det Nominal*
and then rewrite *Nominal* as:	*Det Noun*
and finally rewrite these parts-of-speech as:	*a flight*

- This **sequence of rule expansions** is called a **derivation** of the string of words.
- It is common to represent a derivation by a **parse tree** (commonly shown inverted with the root at the top).

Context Free Grammar

- In the parse tree shown in the below Fig, we can say that the node NP dominates all the nodes in the tree (Det, Nom, Noun, a, flight).
- We can say further that it immediately dominates the nodes Det and Nom.



A parse tree for “a flight”.

Context Free Grammar

- The formal language defined by a CFG is the *set of strings* that are *derivable* from the designated start symbol.
- **Each grammar** must have **one designated start symbol**, which is often called **S**.

Context Free Grammar

Example

Let's add a few additional rules to our inventory. The following rule expresses the fact that a sentence can consist of a noun phrase followed by a **verb phrase**:

$$S \rightarrow NP\ VP \quad \text{I prefer a morning flight}$$

A verb phrase in English consists of a verb followed by assorted other things; for example, one kind of verb phrase consists of a verb followed by a noun phrase:

$$VP \rightarrow Verb\ NP \quad \text{prefer a morning flight}$$

Or the verb may be followed by a noun phrase and a prepositional phrase:

$$VP \rightarrow Verb\ NP\ PP \quad \text{leave Boston in the morning}$$

Or the verb phrase may have a verb followed by a prepositional phrase alone:

$$VP \rightarrow Verb\ PP \quad \text{leaving on Thursday}$$

A prepositional phrase generally has a preposition followed by a noun phrase. For example, a common type of prepositional phrase in the ATIS corpus is used to indicate location or direction:

$$PP \rightarrow Preposition\ NP \quad \text{from Los Angeles}$$

Context Free Grammar

Noun $\rightarrow$	*flights* | *breeze* | *trip* | *morning*
Verb $\rightarrow$	*is* | *prefer* | *like* | *need* | *want* | *fly*
Adjective $\rightarrow$	*cheapest* | *non-stop* | *first* | *latest*
	| *other* | *direct*
Pronoun $\rightarrow$	*me* | *I* | *you* | *it*
Proper-Noun $\rightarrow$	*Alaska* | *Baltimore* | *Los Angeles*
	| *Chicago* | *United* | *American*
Determiner $\rightarrow$	*the* | *a* | *an* | *this* | *these* | *that*
Preposition $\rightarrow$	*from* | *to* | *on* | *near*
Conjunction $\rightarrow$	*and* | *or* | *but*

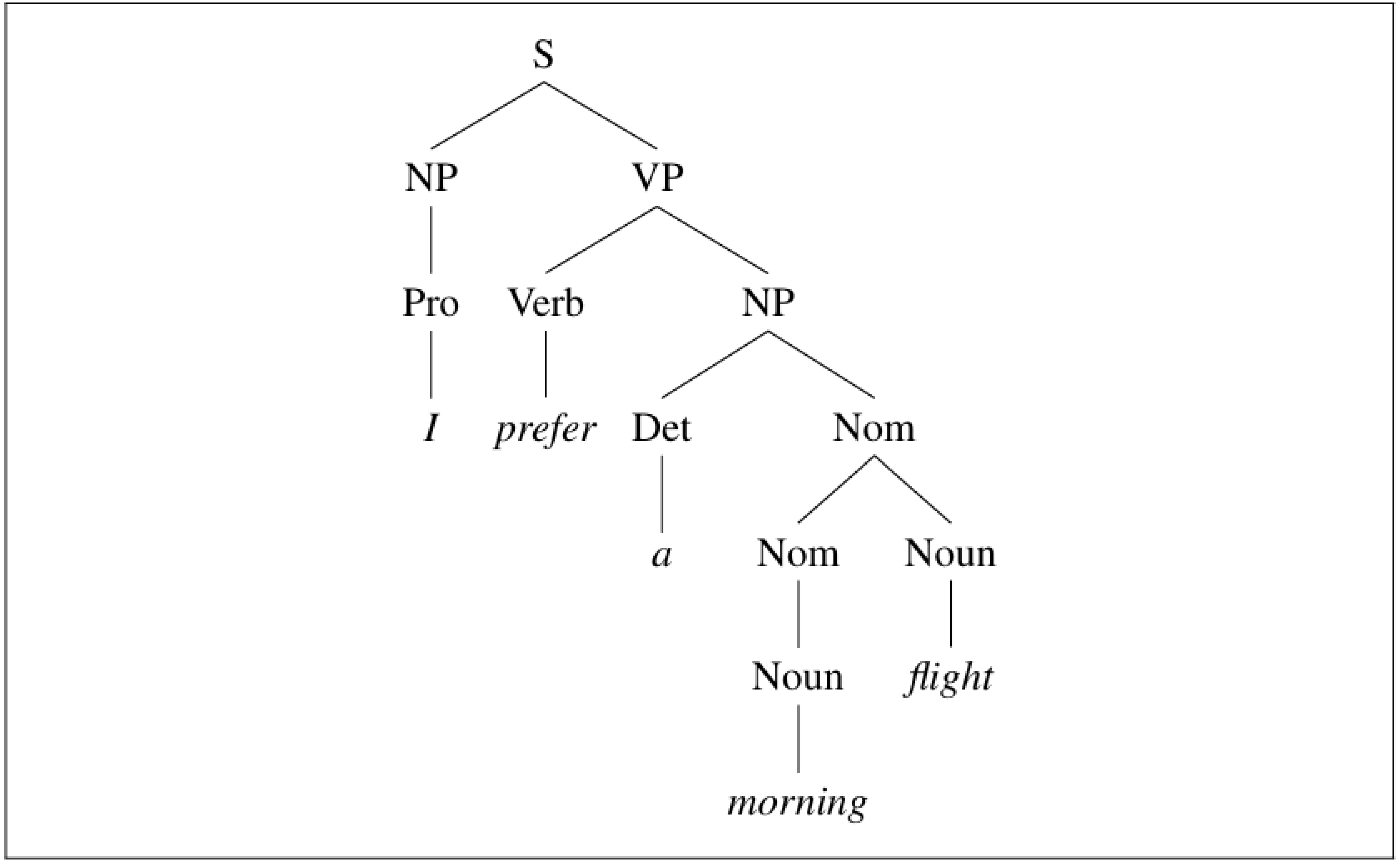
The lexicon for $\mathscr{L}_0$.

Context Free Grammar

Grammar Rules			Examples
S	→	*NP VP*	I + want a morning flight
NP	→	*Pronoun*	I
	|	*Proper-Noun*	Los Angeles
	|	*Det Nominal*	a + flight
Nominal	→	*Nominal Noun*	morning + flight
	|	*Noun*	flights
VP	→	*Verb*	do
	|	*Verb NP*	want + a flight
	|	*Verb NP PP*	leave + Boston + in the morning
	|	*Verb PP*	leaving + on Thursday
PP	→	*Preposition NP*	from + Los Angeles

The grammar for $\mathscr{L}_0$, with example phrases for each rule.

Context Free Grammar



The parse tree for “I prefer a morning flight” according to grammar $\mathscr{L}_0$.

Formal Definition of Context-Free Grammar

- A context-free grammar G is defined by four parameters: N, Σ, R, S (technically this is a “4-tuple”).

N	a set of **non-terminal symbols** (or **variables**)
Σ	a set of **terminal symbols** (disjoint from N)
R	a set of **rules** or productions, each of the form $A \rightarrow \beta$,
	where A is a non-terminal,
	β is a string of symbols from the infinite set of strings $(\Sigma \cup N)*$
S	a designated **start symbol** and a member of N

Context-Free Grammar Parsing

Parsing

Method of analyzing a sentence (string of non-terminals) to determine its structure according to the grammars, to get the meaning out from the sentences.

Set of Rule

↓

CFG

↓

PARSING

↓

Different approaches

Top-Down Parsing (TDP) [start with Start Symbol]

Bottom-up Parsing (BUP) [start from Terminals]

Outputs of Parsing: returns all possible parse trees for the string.

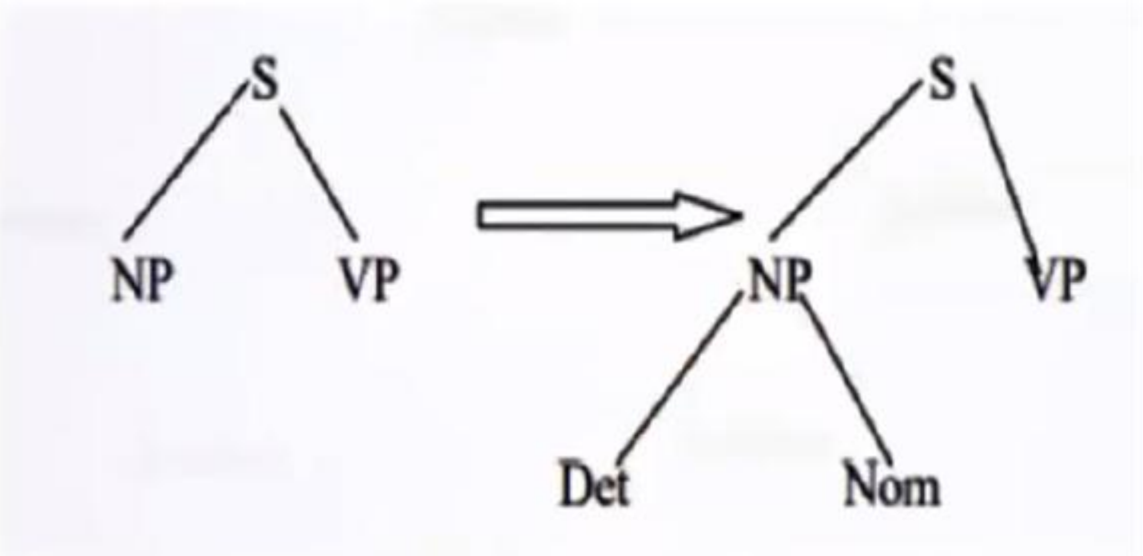
Example (ambiguous sentence): *police are looking for a thief with a bicycle*.

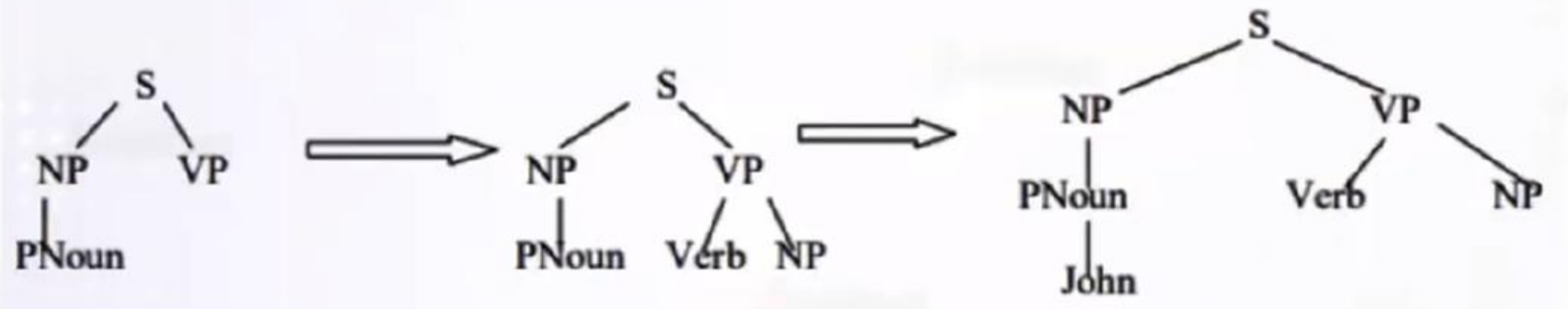
"police with bicycle" or "thief with bicycle"

CFG Top-Down Parsing

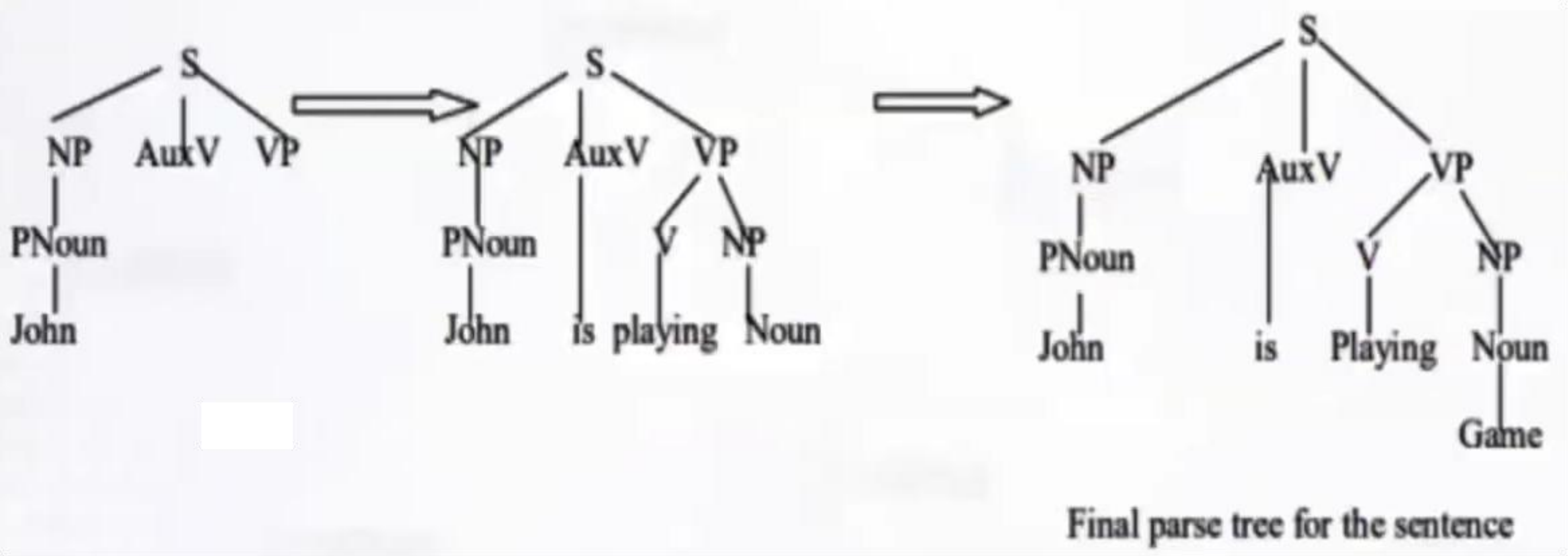
Example 1 : John is playing game

S -> NP VP | NP AuxV VP
NP-> DET Nom | PNoun | Noun
VP->Verb NP | V NP
PNoun-> "John"
AuxV -> is
V->playing
Noun-> Game





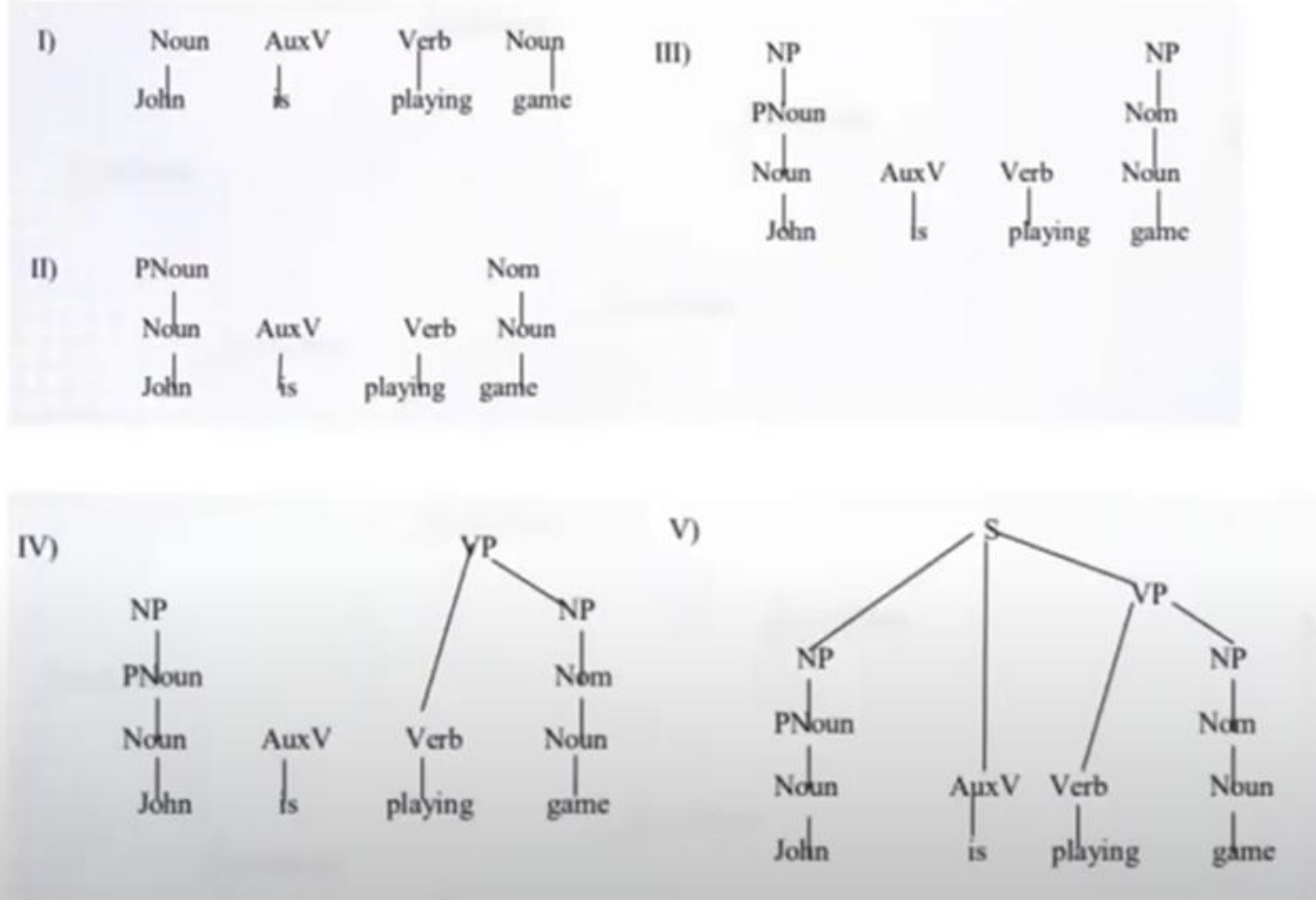
CFG Top-Down Parsing



CFG Bottom-Up Parsing

Example 1: "John is playing game"

S -> NP VP | NP AuxV VP
NP-> DET Nom | PNoun | Nom
VP->Verb NP | V NP
PNoun-> Noun
AuxV -> is
Verb->playing
Noun-> "game" | "John"
Nom->Noun



CFG Top-Down and Bottom-Up Parsing

References

https://www.youtube.com/watch?v=S_t2qDli80s&t=645s

https://www.youtube.com/watch?v=6HF7qUgDxMo